

Purpose of this document

This publication is intended to guide the planning, teaching and assessment of the subject in schools. Subject teachers are the primary audience, although it is expected that teachers will use the guide to inform students and parents about the subject.

This guide can be found on the subject page of the programme resource centre at resources.ibo.org, a password-protected IB website designed to support IB teachers. It can also be purchased from the IB store at store.ibo.org

Additional resources

Additional publications such as specimen papers and markschemes, teacher support materials, subject reports and grade descriptors can also be found on the programme resource centre. Past examination papers as well as markschemes can be purchased from the IB store.

Teachers are encouraged to check the programme resource centre for additional resources created or used by other teachers. Teachers can provide details of useful resources, for example websites, books, videos, journals or teaching ideas.

Acknowledgment

The IB wishes to thank the educators and associated schools for generously contributing time and resources to the production of this guide.

First assessment 2021

The Diploma Programme

The Diploma Programme is a rigorous pre-university course of study designed for students in the 16-19 age range. It is a broad-based two-year course that aims to encourage students to be knowledgeable and inquiring, but also caring and compassionate. There is a strong emphasis on encouraging students to develop intercultural understanding, open-mindedness, and the attitudes necessary for them to respect and evaluate a range of points of view.

The Diploma Programme model

The course is presented as six academic areas enclosing a central core (see figure 1). It encourages the concurrent study of a broad range of academic areas. Students study two modern languages (or a modern language and a classical language), a humanities or social science subject, an experimental science, mathematics and one of the creative arts. It is this comprehensive range of subjects that makes the Diploma Programme a demanding course of study designed to prepare students effectively for university entrance. In each of the academic areas students have flexibility in making their choices, which means they can choose subjects that particularly interest them and that they may wish to study further at university.

Figure 1

The Diploma Programme model



Choosing the right combination

Students are required to choose one subject from each of the six academic areas, although they can, instead of an arts subject, choose two subjects from another area. Normally, three subjects (and not more than four) are taken at higher level (HL), and the others are taken at standard level (SL). The IB recommends 240 teaching hours for HL subjects and 150 hours for SL. Subjects at HL are studied in greater depth and breadth than at SL.

At both levels many skills are developed, especially those of critical thinking and analysis. At the end of the course, students' abilities are measured by means of external assessment. Many subjects contain some element of coursework assessed by teachers.

The core of the Diploma Programme model

All Diploma Programme (DP) students participate in the three course elements that make up the core of the model.

Theory of knowledge (TOK) is a course that is fundamentally about critical thinking and inquiry into the process of knowing rather than about learning a specific body of knowledge. The TOK course examines the nature of knowledge and how we know what we claim to know. It does this by encouraging students to analyse knowledge claims and explore questions about the construction of knowledge. The task of TOK is to emphasize connections between areas of shared knowledge and link them to personal knowledge in such a way that an individual becomes more aware of his or her own perspectives and how they might differ from others.

Creativity, activity, service (CAS) is at the heart of the Diploma Programme. The emphasis in CAS is on helping students to develop their own identities, in accordance with the ethical principles embodied in the IB mission statement and the IB learner profile. It involves students in a range of activities alongside their academic studies throughout the Diploma Programme. The three strands of CAS are creativity (arts, and other experiences that involve creative thinking), activity (physical exertion contributing to a healthy lifestyle) and service (an unpaid and voluntary exchange that has a learning benefit for the student). Possibly, more than any other component in the Diploma Programme, CAS contributes to the IB's mission to create a better and more peaceful world through intercultural understanding and respect.

The extended essay, including the world studies extended essay, offers the opportunity for IB students to investigate a topic of special interest, in the form of a 4,000-word piece of independent research. The area of research undertaken is chosen from one of the students' six Diploma Programme subjects, or in the case of the inter-disciplinary World Studies essay, two subjects, and acquaints them with the independent research and writing skills expected at university. This leads to a major piece of formally-presented, structured writing, in which ideas and findings are communicated in a reasoned and coherent manner, appropriate to the subject or subjects chosen. It is intended to promote high-level research and writing skills, intellectual discovery and creativity. An authentic learning experience, it provides students with an opportunity to engage in personal research on a topic of choice, under the guidance of a supervisor.

Approaches to teaching and approaches to learning

Approaches to teaching and learning across the Diploma Programme refers to deliberate strategies, skills and attitudes which permeate the teaching and learning environment. These approaches and tools, intrinsically linked with the learner profile attributes, enhance student learning and assist student preparation for the Diploma Programme assessment and beyond. The aims of approaches to teaching and learning in the Diploma Programme are to:

- empower teachers as teachers of learners as well as teachers of content
- empower teachers to create clearer strategies for facilitating learning experiences in which students are more meaningfully engaged in structured inquiry and greater critical and creative thinking

- promote both the aims of individual subjects (making them more than course aspirations) and linking previously isolated knowledge (concurrency of learning)
- encourage students to develop an explicit variety of skills that will equip them to continue to be actively engaged in learning after they leave school, and to help them not only obtain university admission through better grades but also prepare for success during tertiary education and beyond
- enhance further the coherence and relevance of the students' Diploma Programme experience
- allow schools to identify the distinctive nature of an IB Diploma Programme education, with its blend of idealism and practicality.

The five approaches to learning (developing thinking skills, social skills, communication skills, self-management skills and research skills) along with the six approaches to teaching (teaching that is inquiry-based, conceptually-focused, contextualized, collaborative, differentiated and informed by assessment) encompass the key values and principles that underpin IB pedagogy.

The IB mission statement and the IB learner profile

The Diploma Programme aims to develop in students the knowledge, skills and attitudes they will need to fulfill the aims of the IB, as expressed in the organization's mission statement and the learner profile. Teaching and learning in the Diploma Programme represent the reality in daily practice of the organization's educational philosophy.

Academic honesty

Academic honesty in the Diploma Programme is a set of values and behaviours informed by the attributes of the learner profile. In teaching, learning and assessment, academic honesty serves to promote personal integrity, engender respect for the integrity of others and their work, and ensure that all students have an equal opportunity to demonstrate the knowledge and skills they acquire during their studies.

All coursework—including work submitted for assessment—is to be authentic, based on the student's individual and original ideas with the ideas and work of others fully acknowledged. Assessment tasks that require teachers to provide guidance to students or that require students to work collaboratively must be completed in full compliance with the detailed guidelines provided by the IB for the relevant subjects.

For further information on academic honesty in the IB and the Diploma Programme, please consult the IB publications *Academic honesty in the IB educational context*, *Effective citing and referencing*, *The Diploma Programme: From principles into practice* and *General regulations: Diploma Programme*. Specific information regarding academic honesty as it pertains to external and internal assessment components of this Diploma Programme subject can be found in this guide.

Acknowledging the ideas or work of another person

Coordinators and teachers are reminded that candidates must acknowledge all sources used in work submitted for assessment. The following is intended as a clarification of this requirement.

Diploma Programme candidates submit work for assessment in a variety of media that may include audio-visual material, text, graphs, images and/or data published in print or electronic sources. If a candidate uses the work or ideas of another person, the candidate must acknowledge the source using a standard style of referencing in a consistent manner. A candidate's failure to acknowledge a source will be investigated by the IB as a potential breach of regulations that may result in a penalty imposed by the IB final award committee.

The IB does not prescribe which style(s) of referencing or in-text citation should be used by candidates; this is left to the discretion of appropriate faculty/staff in the candidate's school. The wide range of subjects, three response languages and the diversity of referencing styles make it impractical and restrictive to insist on particular styles. In practice, certain styles may prove most commonly used, but schools are free to

choose a style that is appropriate for the subject concerned and the language in which candidates' work is written. Regardless of the reference style adopted by the school for a given subject, it is expected that the minimum information given includes: name of author, date of publication, title of source, and page numbers as applicable.

Candidates are expected to use a standard style and use it consistently so that credit is given to all sources used, including sources that have been paraphrased or summarized. When writing text, candidates must clearly distinguish between their words and those of others by the use of quotation marks (or other method, such as indentation) followed by an appropriate citation that denotes an entry in the bibliography. If an electronic source is cited, the date of access must be indicated. Candidates are not expected to show faultless expertise in referencing, but are expected to demonstrate that all sources have been acknowledged. Candidates must be advised that audio-visual material, text, graphs, images and/or data published in print or in electronic sources that is not their own must also attribute the source. Again, an appropriate style of referencing/citation must be used.

Learning diversity and learning support requirements

Schools must ensure that equal access arrangements and reasonable adjustments are provided to candidates with learning support requirements that are in line with the IB documents *Access and inclusion policy* and *Learning diversity and inclusion in IB programmes*.

The documents *Meeting student learning diversity in the classroom* and *The IB guide to inclusive education: a resource for whole school development* are available to support schools in the ongoing process of increasing access and engagement by removing barriers to learning.

Nature of Mathematics

Introduction

Mathematics has been described as the study of structure, order and relation that has evolved from the practices of counting, measuring and describing objects. Mathematics provides a unique language to describe, explore and communicate the nature of the world we live in as well as being a constantly building body of knowledge and truth in itself that is distinctive in its certainty. These two aspects of mathematics, a discipline that is studied for its intrinsic pleasure and a means to explore and understand the world we live in, are both separate yet closely linked.

Mathematics is driven by abstract concepts and generalization. This mathematics is drawn out of ideas, and develops through linking these ideas and developing new ones. These mathematical ideas may have no immediate practical application. Doing such mathematics is about digging deeper to increase mathematical knowledge and truth. The new knowledge is presented in the form of theorems that have been built from axioms and logical mathematical arguments and a theorem is only accepted as true when it has been proven. The body of knowledge that makes up mathematics is not fixed; it has grown during human history and is growing at an increasing rate.

The side of mathematics that is based on describing our world and solving practical problems is often carried out in the context of another area of study. Mathematics is used in a diverse range of disciplines as both a language and a tool to explore the universe; alongside this its applications include analyzing trends, making predictions, quantifying risk, exploring relationships and interdependence.

While these two different facets of mathematics may seem separate, they are often deeply connected. When mathematics is developed, history has taught us that a seemingly obscure, abstract mathematical theorem or fact may in time be highly significant. On the other hand, much mathematics is developed in response to the needs of other disciplines.

The two mathematics courses available to Diploma Programme (DP) students express both the differences that exist in mathematics described above and the connections between them. These two courses might approach mathematics from different perspectives, but they are connected by the same mathematical body of knowledge, ways of thinking and approaches to problems. The differences in the courses may also be related to the types of tools, for instance technology, that are used to solve abstract or practical problems. The next section will describe in more detail the two available courses.

Summary of courses available

Individual students have different needs, aspirations, interests and abilities. For this reason there are two different subjects in mathematics, each available at SL and HL. These courses are designed for different types of students: those who wish to study mathematics as a subject in its own right or to pursue their interests in areas related to mathematics, and those who wish to gain understanding and competence in how mathematics relates to the real world and to other subjects. Each course is designed to meet the needs of a particular group of students. Mathematics: analysis and approaches and Mathematics: applications and interpretation are both offered at SL and HL. Therefore, great care should be taken to select the course and level that is most appropriate for an individual student.

In making this selection, individual students should be advised to take into account the following factors:

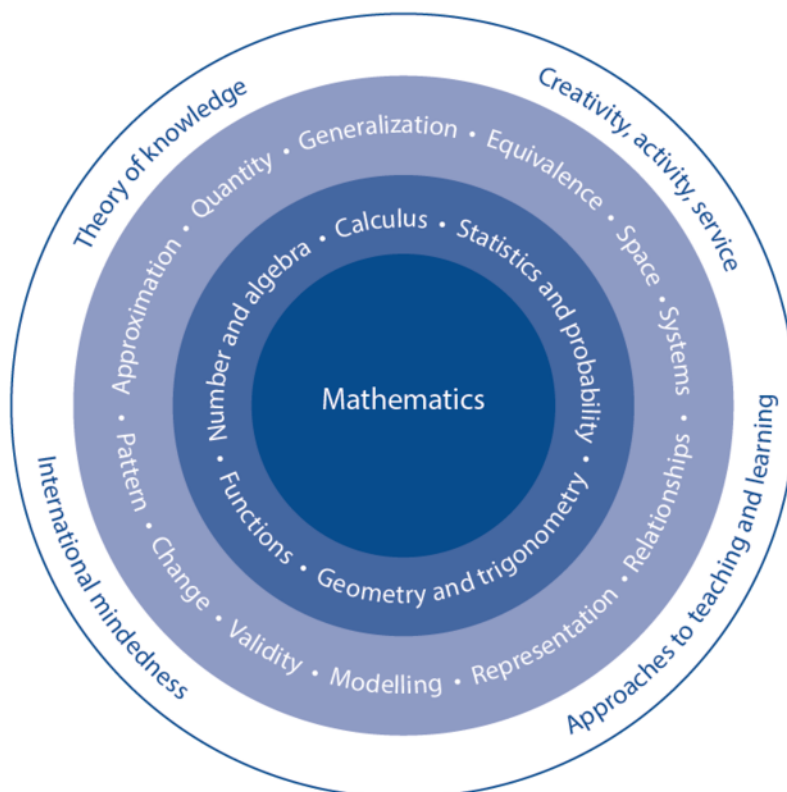
- their own abilities in mathematics and the type of mathematics in which they can be successful
- their own interest in mathematics and those particular areas of the subject that may hold the most interest for them
- their other choices of subjects within the framework of the DP or Career-related Programme (CP)

- their academic plans, in particular the subjects they wish to study in the future
- their choice of career.

Teachers are expected to assist with the selection process and to offer advice to students.

The nature of IB mathematics courses

Figure 2
The mathematics model



The structure of IB DP mathematics courses, with two different routes to choose from, recognizes the two different aspects of mathematics discussed in the introduction.

Mathematics: analysis and approaches is for students who enjoy developing their mathematics to become fluent in the construction of mathematical arguments and develop strong skills in mathematical thinking. They will also be fascinated by exploring real and abstract applications of these ideas, with and without technology. Students who take Mathematics: analysis and approaches will be those who enjoy the thrill of mathematical problem solving and generalization.

Mathematics: applications and interpretation is for students who are interested in developing their mathematics for describing our world and solving practical problems. They will also be interested in harnessing the power of technology alongside exploring mathematical models. Students who take Mathematics: applications and interpretation will be those who enjoy mathematics best when seen in a practical context.

Both subjects are offered at higher level and SL. There are many elements common to both subjects although the approaches may be different. Both subjects will prepare students with the mathematics needed for a range of further educational courses corresponding to the two approaches to mathematics set out above.

Mathematics: analysis and approaches

This course recognizes the need for analytical expertise in a world where innovation is increasingly dependent on a deep understanding of mathematics. This course includes topics that are both traditionally part of a pre-university mathematics course (for example, functions, trigonometry, calculus) as well as topics that are amenable to investigation, conjecture and proof, for instance the study of sequences and series at both SL and HL, and proof by induction at HL.

The course allows the use of technology, as fluency in relevant mathematical software and hand-held technology is important regardless of choice of course. However, Mathematics: analysis and approaches has a strong emphasis on the ability to construct, communicate and justify correct mathematical arguments.

Mathematics: analysis and approaches: Distinction between SL and HL

Students who choose Mathematics: analysis and approaches at SL or HL should be comfortable in the manipulation of algebraic expressions and enjoy the recognition of patterns and understand the mathematical generalization of these patterns. Students who wish to take Mathematics: analysis and approaches at higher level will have strong algebraic skills and the ability to understand simple proof. They will be students who enjoy spending time with problems and get pleasure and satisfaction from solving challenging problems.

Mathematics: applications and interpretation

This course recognizes the increasing role that mathematics and technology play in a diverse range of fields in a data-rich world. As such, it emphasizes the meaning of mathematics in context by focusing on topics that are often used as applications or in mathematical modelling. To give this understanding a firm base, this course also includes topics that are traditionally part of a pre-university mathematics course such as calculus and statistics.

The course makes extensive use of technology to allow students to explore and construct mathematical models. Mathematics: applications and interpretation will develop mathematical thinking, often in the context of a practical problem and using technology to justify conjectures.

Mathematics: applications and interpretation: Distinction between SL and HL

Students who choose Mathematics: applications and interpretation at SL or HL should enjoy seeing mathematics used in real-world contexts and to solve real-world problems. Students who wish to take Mathematics: applications and interpretation at higher level will have good algebraic skills and experience of solving real-world problems. They will be students who get pleasure and satisfaction when exploring challenging problems and who are comfortable to undertake this exploration using technology.

Mathematics and theory of knowledge

The relationship between each subject and theory of knowledge (TOK) is important and fundamental to the DP. The theory of knowledge course provides an opportunity for students to reflect on questions about how knowledge is produced and shared, both in mathematics and also across different areas of knowledge. It encourages students to reflect on their assumptions and biases, helping them to become more aware of their own perspective and the perspectives of others and to become “inquiring, knowledgeable and caring young people” (IB mission statement).

As part of their theory of knowledge course, students are encouraged to explore tensions relating to knowledge in mathematics. As an area of knowledge, mathematics seems to supply a certainty perhaps impossible in other disciplines and in many instances provides us with tools to debate these certainties. This may be related to the “purity” of the subject, something that can sometimes make it seem divorced

from reality. Yet mathematics has also provided important knowledge about the world and the use of mathematics in science and technology has been one of the driving forces for scientific advances.

Despite all its undoubted power for understanding and change, mathematics is in the end a puzzling phenomenon. A fundamental question for all knowers is whether mathematical knowledge really exists independently of our thinking about it. Is it there, “waiting to be discovered”, or is it a human creation? Indeed, the philosophy of mathematics is an area of study in its own right.

Students’ attention should be drawn to questions relating theory of knowledge (TOK) and mathematics, and they should also be encouraged to raise such questions themselves in both their mathematics and TOK classes. Examples of issues relating to TOK are given in the “Connections” sections of the syllabus. Further suggestions for making links to TOK can also be found in the mathematics section of the *Theory of knowledge guide*.

Mathematics and international-mindedness

International-mindedness is a complex and multi-faceted concept that refers to a way of thinking, being and acting characterized by an openness to the world and a recognition of our deep interconnectedness to others.

Despite recent advances in the development of information and communication technologies, the global exchange of mathematical information and ideas is not a new phenomenon and has been essential to the progress of mathematics. Indeed, many of the foundations of modern mathematics were laid many centuries ago by diverse civilisations – Arabic, Greek, Indian and Chinese among others.

Mathematics can in some ways be seen as an international language and, apart from slightly differing notation, mathematicians from around the world can communicate effectively within their field. Mathematics can transcend politics, religion and nationality, and throughout history great civilizations have owed their success in part to their mathematicians being able to create and maintain complex social and architectural structures. Politics has dominated the development of mathematics, to develop ballistics, navigation and trade, and land ownership, often influenced by governments and leaders. Many early mathematicians were political and military advisers and today mathematicians are integral members of teams who advise governments on where money and resources should be allocated.

Science and technology are of significant importance in today’s world. As the language of science, mathematics is an essential component of most technological innovation and underpins developments in science and technology, although the contribution of mathematics may not always be visible. Examples of this include the role of the binary number system, matrix algebra, network theory and probability theory in the digital revolution, or the use of mathematical simulations to predict future climate change or spread of disease. These examples highlight the key role mathematics can play in transforming the world around us.

One way of fostering international-mindedness is to provide opportunities for inquiry into a range of local and global issues and ideas. Many international organisations and bodies now exist to promote mathematics, and students are encouraged to access the resources and often-extensive websites of such mathematical organisations. This can enhance their appreciation of the international dimension of mathematics, as well as providing opportunities to engage with global issues surrounding the subject.

Examples of links relating to international-mindedness are given in the “Connections” sections of the syllabus.

Mathematics and creativity, activity, service

CAS experiences can be associated with each of the subject groups of the DP.

CAS and mathematics can complement each other in a number of ways. Mathematical knowledge provides an important key to understanding the world in which we live, and the mathematical skills and techniques students learn in the mathematics courses will allow them to evaluate the world around them which will help them to develop, plan and deliver CAS experiences or projects.

An important aspect of the mathematics courses is that students develop the ability to systematically analyse situations and can recognize the impact that mathematics can have on the world around them. An

awareness of how mathematics can be used to represent the truth enables students to reflect critically on the information that societies are given or generate, and how this influences the allocation of resources or the choices that people make. This systematic analysis and critical reflection when problem solving may be inspiring springboards for CAS projects.

Students may also draw on their CAS experiences to enrich their involvement in mathematics both within and outside the classroom, and mathematics teachers can assist students in making links between their subjects and students' CAS experiences where appropriate. Purposeful discussion about real CAS experiences and projects will help students to make these links.

The challenge and enjoyment of CAS can often have a profound effect on mathematics students, who might choose, for example, to engage with CAS in the following ways:

- plan, write and implement a "mathematics scavenger hunt" where younger students tour the school answering interesting mathematics questions as part of their introduction to a new school
- as a CAS project students could plan and carry out a survey, create a database and analyse the results, and make suggestions to resolve a problem in the students' local area. This might be, for example, surveying the availability of fresh fruit and vegetables within a community, preparing an action plan with suggestions of how to increase availability or access, and presenting this to a local charity or community group
- taking an element of world culture that interests students and designing a miniature Earth (if the world were 100 people) to express the trend(s) numerically.

It is important to note that a CAS experience can be a single event or may be an extended series of events. However, CAS experiences must be distinct from, and may not be included or used in, the student's Diploma course requirements.

Additional suggestions on the links between DP subjects and CAS can be found in the *Creativity, activity, service teacher support material*.

Prior learning

It is expected that most students embarking on a DP mathematics course will have studied mathematics for at least 10 years. There will be a great variety of topics studied, and differing approaches to teaching and learning. Thus, students will have a wide variety of skills and knowledge when they start their mathematics course. Most will have some background in arithmetic, algebra, geometry, trigonometry, probability and statistics. Some will be familiar with an inquiry approach, and may have had an opportunity to complete an extended piece of work in mathematics.

At the beginning of the syllabus section there is a list of topics that are considered to be prior learning for the mathematics courses. It is recognized that this may contain topics that are unfamiliar to some students, but it is anticipated that there may be other topics in the syllabus itself which these students have already encountered. Teachers should be informed by their assessment of students' prior learning to help plan their teaching so that topics mentioned that are unfamiliar to their students can be incorporated.

Links to the Middle Years Programme

The MYP mathematics framework is designed to prepare students for the study of DP mathematics courses. As students progress from the MYP to the DP or the Career-related Programme (CP) they continue to develop their mathematics skills and knowledge, which will then allow them to go on to study a wide range of topics. Inquiry-based learning is central to mathematics courses in both MYP and the DP, providing students with opportunities to independently and collaboratively investigate, problem solve and communicate their mathematics with an increasing level of sophistication.

MYP mathematics courses are concept-driven, aimed at helping the learner to construct meaning through improved critical thinking and the transfer of knowledge. The MYP courses use a framework of key concepts with which the concepts in the DP mathematics courses are aligned. These concepts are broad, organizing, powerful ideas that have relevance within the subject but also transcend it, having relevance in other

subject groups. The fundamental concepts of MYP mathematics provide a very useful foundation for students following DP mathematics courses.

The aims of the MYP mathematics courses align very closely with those of the DP mathematics courses. The prior learning topics for the DP mathematics courses have been written in conjunction with the MYP mathematics guide.

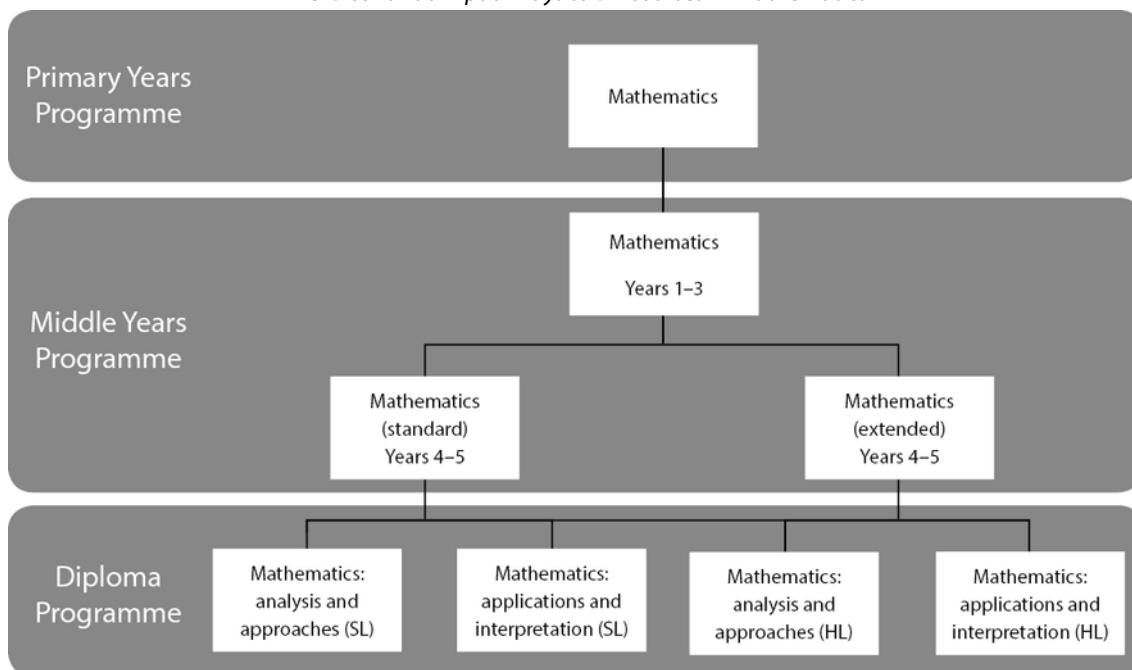
The MYP mathematics assessment objectives and criteria have been developed with both the internal and external assessment requirements of the DP in mind. MYP mathematics students are required to practise and develop their investigation skills, one of the MYP four assessment objectives, giving an important foundation for the internal assessment component of the DP mathematics courses. The MYP assessment objective of thinking critically also corresponds to the higher-order assessment objectives of communication and interpretation, and of reasoning that are expected from a DP mathematics student.

MYP and DP mathematics courses emphasized the use of technology as a powerful tool for learning, applying and communicating mathematics.

Where students in the MYP may select either standard or extended mathematics, at DP there are two mathematics subjects both available at SL and HL. MYP students enrolled in extended mathematics generally elect to take one of the HL mathematics courses in the DP. Students in MYP standard mathematics should seek the recommendation of their teacher when deciding which SL or HL course they are best suited to pursue.

Figure 3

The IB continuum pathways to DP courses in mathematics



Links to the Career-related Programme

In the IB Career-related Programme (CP) students study at least two DP subjects, a core consisting of four components and a career-related study, which is determined by the local context and aligned with student needs. The CP has been designed to add value to the student's career-related studies. This provides the context for the choice of DP courses. Courses can be chosen from any group of the DP. It is also possible to study more than one course from the same group (for example, visual arts and film).

Mathematics may be a beneficial choice for CP students considering careers in, for example, finance, planning, healthcare systems or coding, tourism industries, the technology industry, social informatics, or urban planning. Mathematics helps students to understand the value of systematic approaches, how to analyse complex real-world contexts, how to communicate this concisely and precisely and understand the implications of conclusions.

Mathematics encourages the development of strong written, verbal, and graphical communication skills; critical and complex thinking; and moral and ethical considerations influenced by mathematics that will assist students in preparing for the future global workplace. This in turn fosters the IB learner profile attributes that are transferable to the entire CP, providing relevance and support for the student's learning.

For the CP students, DP courses can be studied at SL or HL. Schools can explore opportunities to integrate CP students with DP students.

Approaches to the teaching and learning of mathematics: applications and interpretation

Conceptual understanding

Concepts are broad, powerful, organizing ideas, the significance of which goes beyond particular origins, subject matter or place in time. Concepts represent the vehicle for students' inquiry into issues and ideas of personal, local and global significance, providing the means by which they can explore the essence of mathematics.

Concepts play an important role in mathematics, helping students and teachers to think with increasing complexity as they organize and relate facts and topics. Students use conceptual understandings as they solve problems, analyse issues and evaluate decisions that can have an impact on themselves, their communities and the wider world.

In DP mathematics courses, conceptual understandings are key to promoting deep learning. The course identifies twelve fundamental concepts which relate with varying emphasis to each of the five topics. Teachers may identify and develop additional concepts to meet local circumstances and national or state curriculum requirements. Teachers can use these concepts to develop connections throughout the curriculum.

Each topic in this guide begins by stating the essential understanding of the topic and highlighting relevant concepts fundamental to the topic. This is followed by suggested conceptual understandings relevant to the content within the topic, although this list is not intended to be prescriptive or exhaustive.

The concepts

Concepts promote the development of a broad, balanced, conceptual and connected curriculum. They represent big ideas that are relevant and facilitate connections within topics, across topics and also to other subjects within the DP.

The twelve concepts identified below support conceptual understanding, can inform units of work and can help to organize teaching and learning. Explanations of each of these concepts in a mathematical context have also been provided.

Approximation	This concept refers to a quantity or a representation which is nearly but not exactly correct.
Change	This concept refers to a variation in size, amount or behaviour.
Equivalence	This concept refers to the state of being identically equal or interchangeable, applied to statements, quantities or expressions.
Generalization	This concept refers to a general statement made on the basis of specific examples.
Modelling	This concept refers to the way in which mathematics can be used to represent the real world.
Patterns	This concept refers to the underlying order, regularity or predictability of the elements of a mathematical system.
Quantity	This concept refers to an amount or number.
Relationships	This concept refers to the connection between quantities, properties or concepts; these connections may be expressed as models, rules or statements. Relationships provide opportunities for students to explore patterns in the world around them.

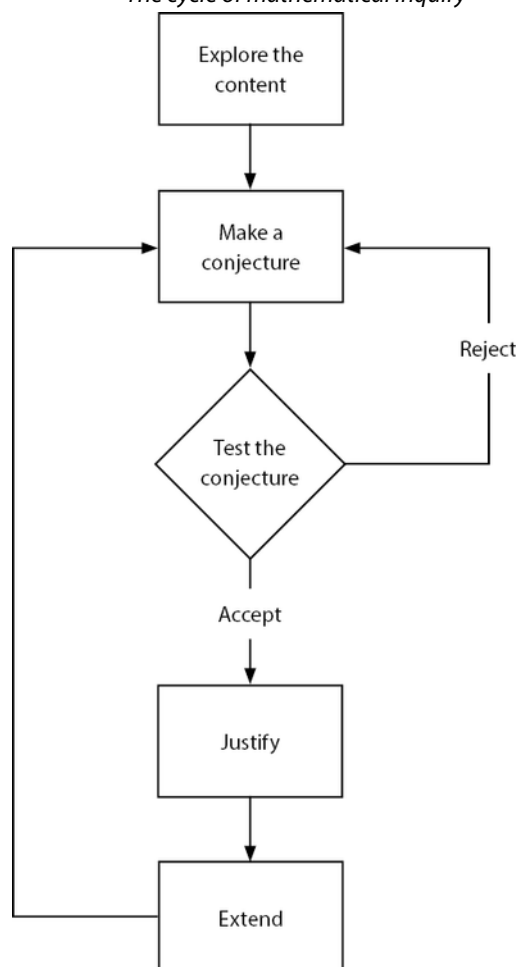
Representation	This concept refers to using words, formulae, diagrams, tables, charts, graphs and models to represent mathematical information.
Space	This concept refers to the frame of geometrical dimensions describing an entity.
Systems	This concept refers to groups of interrelated elements.
Validity	This concept refers to using well-founded, logical mathematics to come to a true and accurate conclusion or a reasonable interpretation of results.

Mathematical inquiry

Approaches to teaching and learning in the DP refer to deliberate strategies, skills and attitudes that permeate the teaching and learning environment. These approaches and tools are intrinsically linked to the IB learner profile, which encourages learning by experimentation, questioning and discovery.

In the IB classroom, students should regularly learn mathematics by being active participants in learning activities. Teachers should therefore provide students with regular opportunities to learn through mathematical inquiry, by making frequent use of strategies which stimulate students' critical thinking and problem-solving skills.

Figure 4
The cycle of mathematical inquiry



Mathematical modelling

Mathematical modelling is an important technique used in problem solving, to make sense of the real world. It is often used to help us better understand a situation, to check the effects of change or to inform decision-making. Engaging students in the mathematical modelling process provides them with such opportunities. It is one of the most useful mathematical skills that students will need to be successful in many non-mathematical as well as mathematical courses and careers.

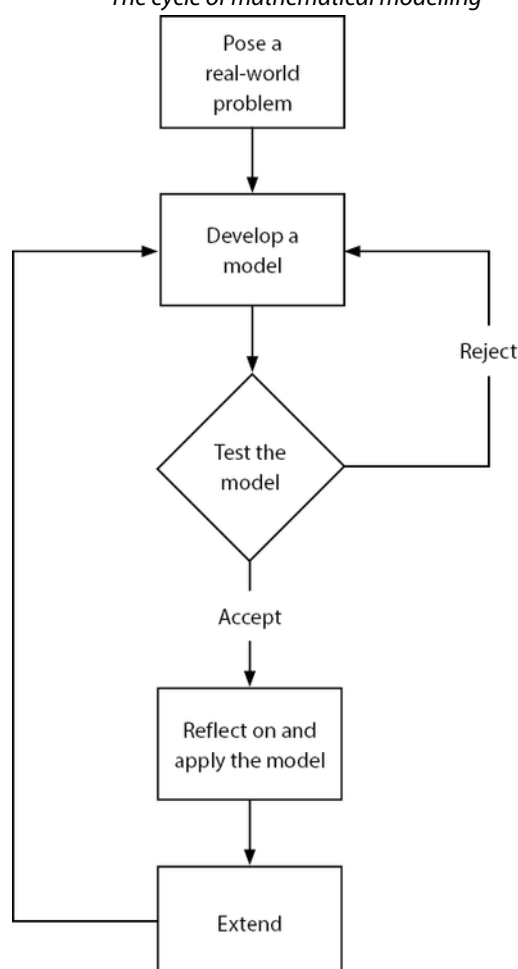
The mathematical modelling process begins with consideration of a situation that exists in the real world and is not usually artificially created. At this stage assumptions sometimes have to be made in order to simplify the situation to allow modelling to take place. There is often a fine balance needed between the simplicity and the accuracy of the model.

The first stage involves choosing or fitting a suitable mathematical representation to the context. This representation is tested to evaluate whether or not it returns expected results. The testing stage allows the results returned by the model to be reflected upon and adaptations to be made to the model if necessary. Once a satisfactory model is established it can be applied or used to explain a situation, check the effects of change or to inform decision-making.

The process of mathematical modelling requires critical reflection throughout the process.

More advice and guidance on the process of mathematical modelling is given in the teacher support material. The cycle of mathematical modelling is illustrated below.

Figure 5
The cycle of mathematical modelling



Use of technology

The use of technology is an integral part of DP mathematics courses. Developing an appreciation of how developments in technology and mathematics have influenced each other is one of the aims of the courses and using technology accurately, appropriately and efficiently both to explore new ideas and to solve problems is one of the assessment objectives. Learning how to use different forms of technology is an important skill in mathematics and time has been allowed in each topic of the syllabus and through the “toolkit” in order to do this.

Technology is a powerful tool in mathematics and in recent years increased student and teacher access to this technology has supported and advanced the teaching and learning of mathematics. Discerning use of technology can make more mathematics accessible and motivating to a greater number of students.

Teachers can use technology to support and enhance student understanding in many ways including:

- to bring out teaching points
- to address misconceptions
- to aid visualization
- to enhance understanding of concepts that would otherwise be restricted by lengthy numerical calculations or algebraic manipulation
- to support students in making conjectures and checking generalizations
- to explicitly make the links between different mathematical representations or approaches.

Students can also use technology to engage with the learning process in many ways including the following:

- to develop and enhance their own personal conceptual understanding
- to search for patterns
- to test conjectures or generalizations
- to justify interpretations
- to collaborate on project based work
- to help organize and analyse data.

In the classroom teachers and students can use technology, working individually or collaboratively, to explore mathematical concepts. The key to successful learning of mathematics with technology is the fine balance between the teacher and student use of technology, with carefully chosen use of technology to support the understanding and the communication of the mathematics itself.

Many topics within the DP mathematics courses lend themselves to the use of technology. Graphical calculators, dynamic graphing software, spreadsheets, simulations, apps, dynamic geometry software and interactive whiteboard software are just a few of the many kinds of technology available to support the teaching and learning of mathematics.

Within the guide the term “technology” is used for any form of calculator, hardware or software that may be available in the classroom. The terms “analysis” and “analytic approach” are generally used in the guide to indicate an algebraic approach that may not require the use of technology. It is important to note there will be restrictions on which technology may be used in examinations, which will be detailed in relevant documents.

As teachers tie together the unifying themes of **mathematical inquiry**, **mathematical modelling** and the **use of technology**, they should begin by providing substantial guidance and then gradually encourage students to become more independent as inquirers and thinkers. DP students should learn to become strong communicators through the language of mathematics. Teachers should create a learning environment in which students are comfortable as risk takers and inquiry, conceptual understanding, collaboration and use of technology feature prominently.

For further information on “Approaches to teaching and learning in the Diploma Programme”, please refer to the publication *The Diploma Programme: From principles into practice*. To support teachers, a variety of resources can be found on the programme resource centre and details of workshops for professional development are available on the public website.

Format of the syllabus

The format of the syllabus section of the mathematics guides is the same for each subject and each level. This structure gives prominence and focus to the aspects of teaching and learning, including conceptual understandings, content and enrichment.

There are five topics and within these topics there are sub-topics. The five topics are:

- number and algebra
- functions
- geometry and trigonometry
- probability and statistics
- calculus

Each topic begins with a section on conceptual understandings. Details are given of which of the twelve key concepts could be used to relate to the topic. Essential understandings give details of the overall aims of the topic and then content-specific conceptual understandings give specific details of the aims and purpose of the topic and sub-topics.

Each topic begins with SL content which is common to both Mathematics: analysis and approaches and to Mathematics: applications and interpretation.

Each topic has SL content followed by advanced higher level (AHL) content. Teachers should ensure that they cover all SL content with SL students, and all SL and AHL content with HL students. The topics are structured so that an informal introduction in the common content may be formalized in the SL content and then extended further in the AHL content. For example the extension of the set of numbers for which something is defined could be integers in the common content, positive real numbers in the SL content and all real and complex numbers in the AHL content.

The content of all five topics at the appropriate level must be taught, however not necessarily in the order in which they appear in the guide. Teachers are expected to construct a course of study that addresses the needs of their students and includes, where necessary, the topics noted in prior learning. Guidance on structuring a course is given in the teacher support material.

Each topic has three sections:

Content: The column on the left contains details of the sub-topics to be covered.

Guidance, clarification and syllabus links: The column on the right contains more detailed information on specific sub-topics listed in the content column. This clarifies the content for the examination and highlights where sub-topics relate to other sub-topics within the syllabus.

Connections: Each topic also contains a short section that provides suggestions for further discussion, including real-life examples and ideas for further investigation.

These suggestions are only a guide and are not exhaustive. A downloadable version of these sections is also available, so that additional connections can be added to the ones already suggested by the IB. Potential areas for connections include:

- **Other contexts: Real-life examples**
- **Links to other subjects:** Suggested links to other subjects within the DP. Note that these are correct for the current (2019) published versions of the guides
- **Aim:** Links to the aims of the course
- **International-mindedness:** Suggestions for discussions
- **TOK:** Suggestions for discussions
- **Links to TSM:** Links to the teacher support materials (TSM) in the “in practice” section of the website

- **Link to specimen paper:** Links to specific questions exemplifying how topics might be examined
- **Use of technology:** Suggestions as to how technology can be used in the classroom to enhance understanding
- **Links to websites:** Suggested websites for use in teaching or learning activities
- **Enrichment:** Suggestions for further discussions that may reinforce understanding.

Planning your course

The syllabus as provided in the subject guide is not intended to be a teaching order. Instead it provides detail of what must be covered by the end of the course. A school should develop a scheme of work that best works for its own students. For example, the scheme of work could be developed to match available resources, to take into account prior learning and experience, or in conjunction with other local requirements.

HL teachers may choose to teach the SL content and AHL content at the same time or teach them in a spiral fashion, by teaching the SL content first and revisiting these topics through the delivery of the AHL topics after this.

However the course is planned, adequate time must be provided for examination revision. Time must also be given to students to reflect on their learning experience and their growth as learners.

Time allocation

The recommended teaching time for HL courses is 240 hours and for SL courses is 150 hours. For mathematics courses at both SL and HL, it is expected that 30 hours will be spent on developing inquiry, modelling and investigation skills. This includes up to 15 hours for work on the internal assessment which is called the exploration. The time allocations given in this guide are approximate, and are intended to suggest how the remaining 210 hours for HL and 120 hours for SL allowed for the teaching of the syllabus might be allocated. The exact time spent on each topic depends on a number of factors, including the background knowledge and level of preparedness of each student. Teachers should therefore adjust these timings to correspond to the needs of their students.

The Toolkit

Time has been allocated within the teaching hours for students to undertake the types of activities that mathematicians in the real world undertake and to allow students time to develop the skill of thinking like a mathematician—in other words providing students with a mathematical toolkit which will allow them to approach any type of mathematical problem. Underpinning this are the six pedagogical approaches to teaching and the five approaches to learning which support all IB programmes. This time gives students opportunities in the classroom for undertaking an inquiry-based approach and focusing on conceptual understanding of the content, developing their awareness of mathematics in local and global contexts, gives them opportunities for teamwork and collaboration as well as time to reflect upon their own learning of mathematics.

Students should be encouraged to actively identify skills that they might add to their personal mathematics toolkit. Teachers are encouraged to make explicit where these skills might transfer across areas of mathematics content and allow students to reflect upon where these skills transfer to other subjects the student is studying.

The teacher support material (TSM) contains a section referred to as the “toolkit”. This section contains ideas and resources that teachers can use with their students to encourage the development of mathematical thinking skills. These resources have been developed by teachers for use in their own classrooms and are not exhaustive.

Formulae and the formula booklet

Formulae are only included in this guide document where there may be some ambiguity. All formulae required for the course are in the mathematics formula booklet.

From the beginning of the course it is recommended that teachers ensure students are familiar with the contents of the formula booklet by either giving students a printed copy or making an electronic copy available to them.

Each student is required to have access to a clean copy of the formula booklet during the examination. For each examination, it is the responsibility of the school to download a copy of the formula booklet from IBIS or the programme resource centre, check that there are no printing errors, and ensure that there are sufficient copies available for all students.

Command terms and notation list

Teachers and students need to be familiar with the IB notation list and the command terms, as these will be used without explanation in the examination papers. The “Glossary of command terms” and “Notation list” appear as appendices in this guide.

Aims

The aims of all DP mathematics courses are to enable students to:

1. develop a curiosity and enjoyment of mathematics, and appreciate its elegance and power
2. develop an understanding of the concepts, principles and nature of mathematics
3. communicate mathematics clearly, concisely and confidently in a variety of contexts
4. develop logical and creative thinking, and patience and persistence in problem solving to instil confidence in using mathematics
5. employ and refine their powers of abstraction and generalization
6. take action to apply and transfer skills to alternative situations, to other areas of knowledge and to future developments in their local and global communities
7. appreciate how developments in technology and mathematics influence each other
8. appreciate the moral, social and ethical questions arising from the work of mathematicians and the applications of mathematics
9. appreciate the universality of mathematics and its multicultural, international and historical perspectives
10. appreciate the contribution of mathematics to other disciplines, and as a particular “area of knowledge” in the TOK course
11. develop the ability to reflect critically upon their own work and the work of others
12. independently and collaboratively extend their understanding of mathematics.

Assessment objectives

Problem solving is central to learning mathematics and involves the acquisition of mathematical skills and concepts in a wide range of situations, including non-routine, open-ended and real-world problems. Having followed a DP mathematics course, students will be expected to demonstrate the following:

1. **Knowledge and understanding:** Recall, select and use their knowledge of mathematical facts, concepts and techniques in a variety of familiar and unfamiliar contexts.
2. **Problem solving:** Recall, select and use their knowledge of mathematical skills, results and models in both abstract and real-world contexts to solve problems.
3. **Communication and interpretation:** Transform common realistic contexts into mathematics; comment on the context; sketch or draw mathematical diagrams, graphs or constructions both on paper and using technology; record methods, solutions and conclusions using standardized notation; use appropriate notation and terminology.
4. **Technology:** Use technology accurately, appropriately and efficiently both to explore new ideas and to solve problems.
5. **Reasoning:** Construct mathematical arguments through use of precise statements, logical deduction and inference and by the manipulation of mathematical expressions.
6. **Inquiry approaches:** Investigate unfamiliar situations, both abstract and from the real world, involving organizing and analyzing information, making conjectures, drawing conclusions, and testing their validity.

Assessment objectives in practice

Assessment objectives	Paper 1 %	Paper 2 %	Paper 3 % HL only	Exploration %
Knowledge and understanding	20-30	20-30	10-20	5-15
Problem solving	20-30	20-30	20-30	5-20
Communication and interpretation	20-30	20-30	20-30	15-25
Technology	20-35	20-35	10-30	10-20
Reasoning	5-15	10-20	10-20	5-25
Inquiry approaches	5-15	5-20	15-30	25-35